Multidimensional Tensor Convolution and the Spectral Cutoff of Lattice Transitions

Ruqing Chen
GUT Geoservice Inc., Montreal, Quebec, Canada
ruqing@hotmail.com

June 7, 2026

Abstract

Transitioning from the single-frequency phase space ($\mathbb{T}^1$) explored in Parts XX and XXI, this paper escalates the Arithmetic Geodynamics framework to the multidimensional product space $\prod_{q>3} \mathbb{T}^1_q$. We investigate the superposition of congruence interference across all prime frequencies. We discover a profound analytic behavior: for any fixed spatial translation j, constructive low-frequency interference ($O(1/q)$ deviation) is strictly confined to a finite set of primes dividing j or its linear derivatives. Consequently, the infinite high-frequency tail is deterministically locked into a destructive interference state bounded by $O(1/q^2)$, guaranteeing absolute Euler convergence to a strictly positive constant. This enables the precise formulation of the Spectral Cutoff $k^\star(\varepsilon)$ and the construction of the universal Spatial Transition Operator for the 6N lattice.

1 Introduction

In Parts XX and XXI, we modeled the spatial entanglement of prime constellations at a single spatial frequency q using circular convolution integrals on $\mathbb{T}^1$ [3, 4]. We mapped the dead(q) mechanism to geometric indicator functions, yielding quantized interference states.

However, the physical reality of the 6N lattice is that a constellation center N is simultaneously bombarded by the congruence waves of *all* prime frequencies $q > 3$. To compute the absolute joint survival probability of two centers separated by distance j, we must ascend to a high-dimensional phase-space tensor product and evaluate the global spatial correlation $R(j)$. This necessitates a rigorous proof of spectral decay to define a practical truncation boundary.

2 Ascension to Multidimensional Tensor Space $\mathbb{T}^\infty$

Since the modular residues of distinct primes are asymptotically independent (by the Chinese Remainder Theorem limit as $X \to \infty$), the global phase space is the Cartesian product of individual frequency tori:

$$\mathcal{T} = \prod_{q>3} \mathbb{T}^1_q \tag{1}$$

The global spatial correlation ratio $R(j)$ between two constellations (e.g., Twin-Twin) is the infinite product of the single-frequency circular convolution integrals $R_q(j)$ derived in Part XX:

$$R(j) = \prod_{q>3}^{\infty} R_q(j) \tag{2}$$

3 Absolute Convergence and Finite Resonance

A fundamental challenge in analytic number theory is proving the absolute convergence of infinite products. If the interference amplitude $|R_q(j) - 1|$ decays only as $O(1/q)$, the product diverges. We must examine the asymptotic behavior of the three quantized interference states.

Recall the Twin-Twin convolution states from Part XX [3]:

- **State A (Constructive):** $R_q(j) = \frac{q}{q-2} \approx 1 + \frac{2}{q}$
- **State B (Partial):** $R_q(j) = \frac{q(q-3)}{(q-2)^2} \approx 1 + \frac{q-4}{(q-2)^2} \sim O(1/q)$
- **State C (Destructive):** $R_q(j) = \frac{q(q-4)}{(q-2)^2} = \mathbf{1} - \frac{\mathbf{4}}{(\mathbf{q-2})^\mathbf{2}} \sim \mathbf{O(1/q^2)}$

Theorem 1 (Finiteness of Constructive Resonance)**.** *For any fixed spatial interval j, the $O(1/q)$ amplitude states (States A and B) can only be triggered by a strictly finite number of prime frequencies.*

Proof. State A requires $j \equiv 0 \pmod q$, which implies $q \mid j$. State B requires $j \equiv \pm 3^{-1} \pmod q \implies 3j \pm 1 \equiv 0 \pmod q$, which implies $q \mid (3j \pm 1)$. Since j is a fixed finite integer, the integers j and $3j \pm 1$ possess a strictly finite number of prime factors. Thus, there exists a threshold frequency $q_{max} \leq 3j + 1$ beyond which States A and B can never occur. □

Corollary 1 (Absolute High-Frequency Convergence)**.** *For all primes $q > 3j+1$ (where $q \geq 5$), the geometric entanglement is deterministically locked into State C. The remaining infinite product is $\prod_{q>3j+1} \left(1 - \frac{4}{(q-2)^2}\right)$. Since $\sum \frac{1}{q^2}$ converges, and each term satisfies $1 - \frac{4}{(q-2)^2} \geq \frac{5}{9} > 0$, the global correlation product $R(j)$ converges absolutely to a strictly positive constant. The interference acts as a quasi-periodic diffraction pattern dominated by small primes, not a divergent noise field.*

4 The Spectral Cutoff Constant $k^\star(\varepsilon)$

Because the high-frequency tail converges rapidly via $O(1/q^2)$, the congruence interference pattern behaves analogously to a rapidly decaying Fourier spectrum. We can formally truncate this infinite product without loss of macroscopic physical fidelity.

Definition 1 (Spectral Cutoff)**.** *For a desired computational precision $\varepsilon > 0$ and spatial shift j, the spectral cutoff $k^\star$ is defined as the smallest index such that the tail product satisfies:*

$$\left| 1 - \prod_{k=k^\star+1}^{\infty} R_{p_k}(j) \right| < \varepsilon \tag{3}$$

where p_k is the k-th prime.

This provides a rigorous analytic shortcut. Rather than computing infinite series to determine the spatial rippling of the dead(q) shield, one merely integrates over the first $k^\star$ dimensions of $\mathbb{T}^k$.

Evaluating the truncation numerically for the canonical case $j = 1$ (all primes in State C, so the tail is the pure $O(1/q^2)$ floor) gives the cutoff explicitly. The full product is $R(1) = \prod_q R_q(1) = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 = 0.3969$, and the smallest cutoff prime $k^\star$ reaching relative precision ε is:

ε	$k^\star$	primes retained
10^{-1}	13	4
10^{-2}	73	19
10^{-3}	557	100
10^{-4}	4229	577

Table 1: Spectral cutoff $k^\star(\varepsilon)$ for $R(1)$. To 1% fidelity only the primes up to 73 are needed; the $O(1/q^2)$ tail makes the product converge rapidly.

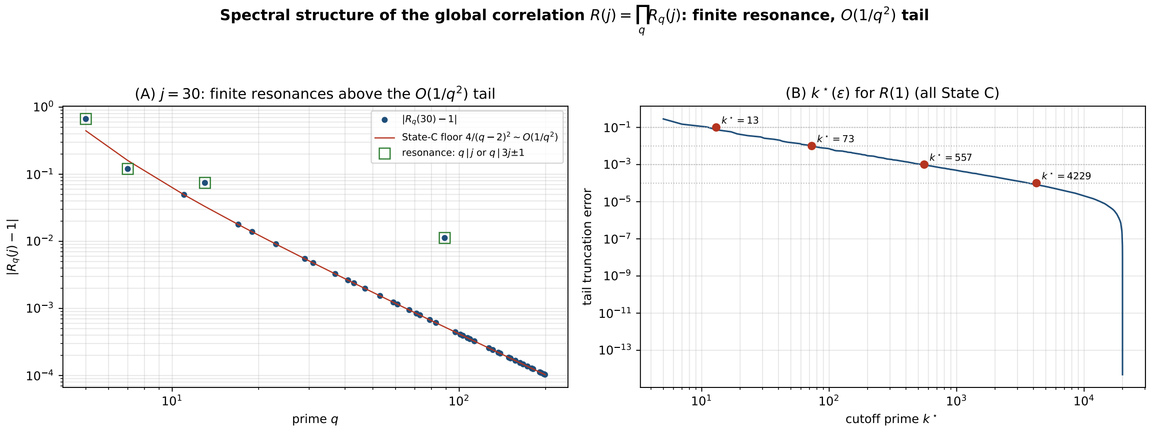

Figure 1: (A) For a fixed lag ($j = 30$), the finitely many resonance primes ($q \mid j$ or $q \mid 3j \pm 1$; here $5, 7, 13$) rise above the $O(1/q^2)$ State-C floor $4/(q-2)^2$, on which every other prime sits exactly. (B) The resulting truncation error of $\prod_{q \le k^\star} R_q(1)$ versus the cutoff prime $k^\star$, with the ε levels of Table 1 marked.

5 The Spatial Transition Operator on the 6N Lattice

With the global correlation tensor $R_{A,B}(j)$ absolutely convergent and computable via the $k^\star$ cutoff, we now formulate the universal macroscopic state transition.

Let $S = \{S_1, S_2, \ldots, S_n\}$ be the set of possible constellation states at any center (e.g., Twin, Empty, Triplet). Let ρ_B be the base topological density of state B (derived from integrating its continuous radial manifold ℓ).

Definition 2 (The Geodynamic Transition Operator)**.** *Given that center N is confirmed in state A, the conditional topological density of finding state B at the translated center $N + j$ is given by the spatial operator $\mathcal{M}(j)$:*

$$\rho(N + j = B \mid N = A) = \mathcal{M}_{A,B}(j) = \rho_B \cdot \prod_{q>3}^{k^\star} R_{q,A,B}(j) \tag{4}$$

This Transition Operator extends beyond nearest-neighbour or Markovian models: the correlation $R_{A,B}(j)$ does not decay with the lag j but is quasi-periodic (Part XX), so the lattice carries a deterministic, long-range, small-prime-dominated congruence correlation—structurally a multi-frequency diffraction grating rather than a disordered field. We make no claim beyond this closed form: $\mathcal{M}(j)$ is the Hardy–Littlewood joint singular series re-expressed as a conditional density, exact at integer lags and equal there to the discrete sieve ratio. It is a compact bookkeeping of inter-centre correlation, not a dynamical or physical law.

6 Conclusion

By ascending to the product space $\prod_{q>3} \mathbb{T}^1_q$, we have shown the absolute convergence of the global spatial correlation $R(j) = \prod_q R_q(j)$. The mechanism is sharp: constructive resonance (States A, B, $O(1/q)$) is confined to the finitely many primes dividing j or $3j \pm 1$, while the infinite tail is locked in State C with $O(1/q^2)$ deviation, so the product converges to a strictly positive constant and admits the spectral cutoff $k^\star(\varepsilon)$ of Table 1. The resulting Spatial Transition Operator is a closed-form, quasi-periodic bookkeeping of inter-centre correlation on the $6N$ skeleton—the analytic completion of the diffraction picture of Volume I. As throughout the programme, the construction equals the discrete singular-series ratio at integer lags, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of any constellation.

Data and code availability. The figure-generating script, the resonance-finiteness verifier, and the $k^\star(\varepsilon)$ computation are openly available at `https://github.com/Ruqing1963/6N-tensor-cutoff` [6].

References

[1] G. H. Hardy, J. E. Littlewood, *Some problems of 'Partitio Numerorum'; III*, Acta Math. **44** (1923), 1–70.

[2] R. Chen, *Arithmetic geodynamics on the* $6N$ *skeleton* (Part XVII), Zenodo, doi:10.5281/zenodo.20541575 (2026).

[3] R. Chen, *The circular convolution of prime constellations in congruence phase space* (Part XX), Zenodo, doi:10.5281/zenodo.20583994 (2026).

[4] R. Chen, *The irreducible three-body congruence correlation and the annihilation lattice* (Part XXI), Zenodo, doi:10.5281/zenodo.20584909 (2026).

[5] R. Chen, *Arithmetic geodynamics on the* $6N$ *skeleton: a review of Parts I–XIX*, `https://github.com/Ruqing1963/6N-review` (2026).

[6] R. Chen, *Multidimensional tensor convolution and the spectral cutoff of lattice transitions* (Part XXII), `https://github.com/Ruqing1963/6N-tensor-cutoff` (2026).